



Online diagnosis and prediction of power battery voltage comprehensive faults for electric vehicles based on multi-parameter characterization and improved K-means method

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ABSTRACT

The battery terminal voltage in the power battery system is a comprehensive indicator of its internal resistance, capacity, state of charge (SoC) and other parameters, which can more comprehensively assess the safety performance of the battery system, so it is of great significance to accurately diagnose and predict the voltage faults of individual cells. Based on this, two-dimensional fault characteristics that can effectively recognize the voltage fluctuation are first extracted. And then based on the improved K-means method to carry out the identification of fault cells. In order to achieve online applications and comprehensive detection of different forms of voltage faults, this paper proposes for the first time a double sliding time window simultaneous implementation strategy and optimizes the window length and evaluation coefficient (EC) threshold based on a data-driven approach, and the proposed algorithm can be performed in real time without any significant delay. Finally, the necessity, reliability and stability of the method are verified and compared with Shannon entropy method and correlation coefficient method. Results indicate the method in this article is capable to recognize the various data patterns of the potential threat and can accurately identify anomalies prior to thermal runaway (TR) or failure of the vehicle.

1. Introduction

1.1. Research motivation

Compared with traditional fuel vehicles, electric vehicles (EVs) have attracted much attention and support due to their ability to effectively reduce environmental pollution and alleviate the energy crisis, as well as their obvious advantages in the fields of driverless and intelligent transportation [1–3]. However, with the explosive growth of the electric vehicle base, vehicle safety accidents have long been commonplace. Among them, the power battery as the core component is considered to be the main culprit, which not only increases people's anxiety about vehicle safety, but also hinders its commercial expansion, hence the increasing research in this area [4,5]. Generally speaking, power battery system faults can often be traced back to specific individual cells, and the fault characteristics can be reflected in the cell voltage in real time, and are mostly manifested as inconsistent fluctuation. Based on this, this paper will be dedicated to finding an efficient and accurate method to realize online diagnosis and prediction of abnormal voltage fluctuation

faults.

1.2. Literature review

At present, the common research technology routes for battery safety of EVs are summarised in Fig. 1.

To improve the safety performance of power battery systems, various related technologies have been developed and applied. Specifically, there are two main ways: one is to reduce the defects of batteries before they leave the factory by continuously improving the manufacturing process and packaging technology [6]. The second is the safety detection during battery operation to minimize the occurrence of safety accidents and the losses caused by them [7].

At the manufacturing stage of batteries, the production process is extremely complex and environmentally demanding, and any small parameter errors may affect the performance of the battery [8,9]. For a long time, a great deal of effort has been devoted to the study of such problems, and a range of achievements have been achieved. For example, the literature [10] for the first time used ceramic aqueous solution and acrylic copolymer solution as the coating solutions to

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Abbreviations		FP	The number of falsely identified faulty cells
Acronyms		IPCC	Improved Pearson correlation coefficient
ACC	Accuracy	ISC	Internal short circuit
BMS	Battery management system	LIBS	Lithium-ion battery systems
BN	Boron nitride	LOF	Local outlier factors
DBSCAN	Density-Based Spatial Clustering of Applications with Noise	LSTM	Long short-term memory neural network
DNPSO	Dynamic neighbourhood particle swarm optimisation	NCM	Li(NiCoMn)O ₂
ECM	Equivalent circuit model	PCC	Pearson correlation coefficient
ECs	Evaluation coefficients	SD	Standard deviation
ESC	External short circuit	SDO	Sparse data observer
EVs	Electric vehicles	SoC	State of charge
FMEA	Fault mode and effects analysis	SVM	Support vector machine
FN	The number of falsely identified normal cells	TN	The number of correctly identified normal cells
		TP	The number of correctly identified faulty cells
		TR	Thermal runaway

produce a new battery multilayer-coated separator, which greatly improving the reliability and safety of batteries for long-term high-energy-density operation. The literature [11] proposed an effective approach for improving the cycling stability of Li metal anode via a facile coating of commercial separator with a thermally conductive boron nitride (BN) nanosheets, the implementation of this method significantly slows down the growth rate of lithium dendrites and the decline rate of the Coulombic efficiency compared to the pristine separator. In addition, the literature [12] discussed in detail two fault mode categories, positioning error and multiple layer handling, and focuses on the application of the established Fault Mode and Effects Analysis (FMEA) method to the process of electrode packing, thus enabling the design and optimisation of large-scale battery production processes.

In addition, real-time safety detection is a necessary guarantee for reliable battery operation. In this regard, a lot of research has been carried out by domestic and international scholars, with research methods focusing on both model-based and model-free approaches. Common power battery models include the equivalent circuit model (ECM) and the electrochemical model, of which the ECM is relatively simple and has acceptable model accuracy, and is therefore widely used. For example, Li et al. [13] proposed a novel battery fault diagnosis method combining the long short-term memory (LSTM) recurrent neural network and the ECM. Feng et al. [14] presented a model-based fault

diagnosis algorithm for online internal short circuit (ISC) detection. Chen et al. [15] used an improved first-order RC model and the Dynamic Neighbourhood Particle Swarm Optimisation (DNPSO) algorithm to achieve the diagnosis of external short circuit (ESC) fault. There is no doubt that many advanced fault diagnosis algorithms have been proposed by model-based methods, but the validation of the algorithms is mostly carried out based on static battery experimental data, which tends to have poor diagnostic accuracy for real-world operating EVs. In addition, coupled with the limitations of model parameter accuracy, data quality, and computational complexity, there are still some limitations to their scope of application.

Model-free methods are widely used because they do not require the construction of battery mechanism models and complex algorithmic models, which mainly contain statistical, machine learning and signal processing methods. With the aid of statistical theory, Ma et al. [16] proposed a connection fault detection method for series-connected lithium-ion battery systems (LIBS) based on statistical analysis. Zhao et al. [17] presented a battery fault and defect detection method for EVs based on big data statistics, and Sun et al. [18] proposed a three-layer fault detection strategy for LIBS based on statistical methods. It is well known that statistical methods are simple and effective when the test data conforms to a certain statistical distribution, but for real-world EVs, the differences between different cells are extremely random, and

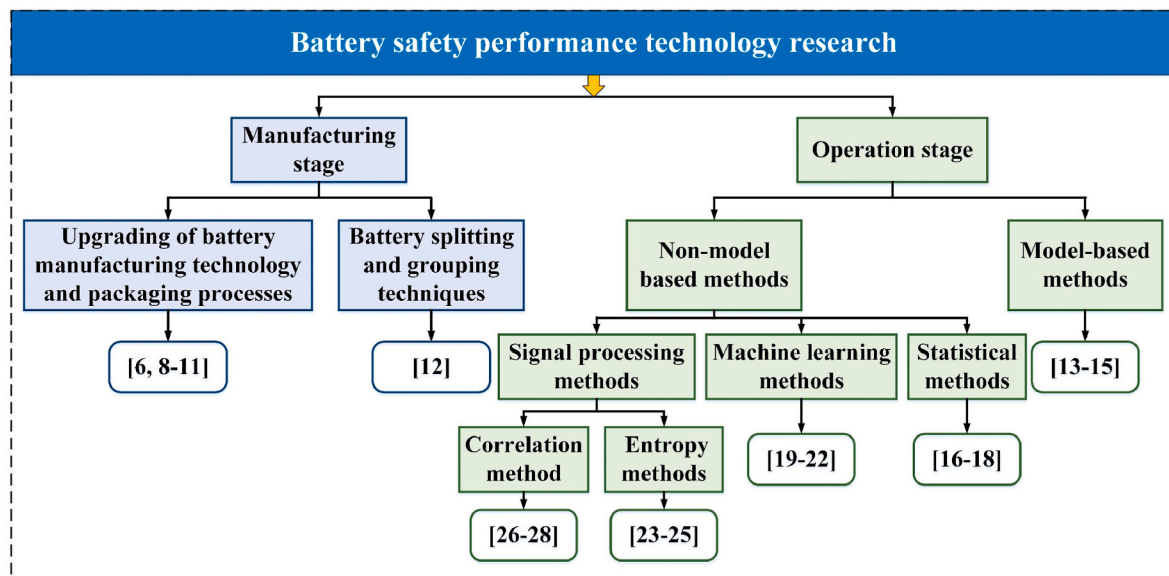


Fig. 1. Battery safety detection classification.

parameters such as cell voltage and internal resistance do not strictly obey the Gaussian distribution. Therefore, some normal cells are often misdiagnosed as faulty cells when statistical methods are forced to be used for fault detection, making it difficult to guarantee accuracy [19].

The measurement signals of faulty cells in the battery pack usually have outlier characteristics. Therefore, outlier detection algorithms in machine learning methods are widely used in the fields of power battery fault diagnosis and thermal runaway (TR) prediction. For example, Local Outlier Factor (LOF) [20], Support Vector Machine (SVM) [21], Sparse Data Observer (SDO) [22], Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [19], and so on. Such methods can not only diagnose the time and location of battery fault, but also realize the assessment of the degree of fault, but the selection of hyperparameters in the algorithm requires manual intervention to complete, and the computational cost will increase exponentially if improperly selected. Therefore, the computational efficiency and the adaptability of the parameters of the machine learning algorithms still deserve further research.

Signal processing based methods mainly analyze the waveform and amplitude of the measured signal in the time or frequency domain. Commonly measured signals include voltage signals, current signals, and temperature signals. The entropy method of signal processing is widely used to diagnose abnormal voltage fluctuations because it can assess the degree of disorder in the system. For example, Wang et al. [23] proposed an online diagnosis method for voltage fault based on improved Shannon entropy and Z-score. Shang et al. [24] proposed a method for diagnosing and isolating early short-circuit and open-circuit faults in batteries based on modified sample entropy, and Hong et al. [25] presented a collaborative battery multi-fault diagnosis method based on modified multiscale entropy. In addition to the entropy method, the correlation coefficient is also favored in the field of battery fault diagnosis due to its inherent mathematical nature that ensures the robustness of the method in practice when there is ageing or imbalance in the battery pack. For example, Xia et al. [26] presented a short-circuit fault detection method based on the correlation coefficient of voltage curves. Kang et al. [27] proposed a multi-fault diagnosis strategy based on a redundant interleaved voltage measurement topology and a correlation coefficient method. Further, Kang et al. [28] achieved the identification and isolation of sensor fault and battery fault by changing the voltage measurement topology without adding additional voltage sensors.

1.3. Contributions of this work

This paper is dedicated to finding an accurate and fast online diagnosis and prediction method for abnormal voltage fluctuation faults, and the main contributions can be summarised as follows:

- (1) The standard deviation (SD) and the improved Pearson correlation coefficient (IPCC) are extracted as two-dimensional fault features characterising voltage fluctuations, and the two features can achieve accurate fault diagnosis together.
- (2) The computational efficiency is greatly improved by the fact that we use a sliding time window to process the data and select the cell with the maximum risk of failure as the initial clustering center for the K-means method.
- (3) To achieve comprehensive and effective detection of voltage faults, we propose for the first time a dual sliding time window simultaneous implementation strategy and use a data-driven approach to optimize the window length and EC threshold, which allows the proposed algorithm to be executed in real time with high sensitivity.
- (4) The necessity, reliability and stability of the method are verified using real-world EVs data with different fault forms. The results show that the method can accurately predict faults before vehicle

fault alarms or TR. The average accuracy of the method exceeds 98%.

1.4. Organization of the paper

The rest of the paper is structured as follows. Section 2 describes the object of the research. Section 3 presents the specific diagnosis method for abnormal voltage fluctuation faults. Section 4 describes the selection process of the key parameters. In Section 5, the results of the real application are analyzed in detail, and finally, the key conclusions are summarised in Section 6.

2. Research object

The real-world vehicle data involved in this research are from a company's EVs big data platform. The data of seven vehicles with abnormal voltage fluctuations or TR and three normal vehicles are retrieved from the platform. All of the batteries are Li(NiCoMn)O₂ (NCM) batteries and connected in series, which include one type of commercial vehicles (model 1) and one type of passenger vehicles (model 2), and the battery pack layout of the studied vehicles can be illustrated as Fig. 2. The main information of vehicles and batteries are summarised in Table 1.

The model is discussed and analyzed based on the voltage data on the day of the accident of faulty vehicle No.1~No.7. The summary results of the voltage curves are shown in Fig. 3. Vehicles No.1, No.3, and No.5~No.7 have TR incidents. No.2 vehicle has a sudden voltage drop fault. No.4 vehicle has an excessive voltage difference alarm message, and vehicle No.8~No.10 are normal vehicles. Specifically, vehicles No.1 and No.5 experienced TR in Cell 21 and Cell 92, respectively, near the end of charging. In No.2 vehicle, the one-fault takes place and resulted in a sudden voltage drop of 80mV for Cell 131 voltage curve at the 310-th sampling point, but the battery voltage gets recovered rapidly after one-off fault due to self-repair and internal circuit equalization. Vehicles No.3 and No.7 showed TR during discharge of Cell 150 and Cell 151, and Cell 12, respectively. No.3 vehicle is identified by professionals to confirm that the TR is caused by a bump on the battery pack, and Fig. 3 (c) clearly shows a sudden drop in voltage for Cell 150 and Cell 151 at the 1347-th sampling point, then recovered rapidly, but the voltage for Cell 151 gradually showed outlier characteristics until the final TR at the 1383rd frame. There is a voltage inconsistency in Cell 34 in No.4 vehicle and the degree of inconsistency gradually increases over time until the maximum voltage difference exceeds a 200mV threshold at the 4189th frame thereby triggering the voltage fault alarm. No.6 vehicle TR occurs in the driving process after a full charge of approximately 1 h, and no fault alarm information is recorded before TR. Due to the settings of the battery management system (BMS), the measured out-of-normal range voltage is recorded as 5V to indicate the error state. Therefore, the voltage readings for Cell 150 and Cell 65 in Fig. 3 (c) and (f) are marked as 5V due to a sudden drop beyond the threshold, respectively. There is a voltage inconsistency for Cell 32 in No.10 vehicle. According to the fault characteristics, voltage fluctuation faults can be divided into two forms: progressive fluctuation fault (consistent trend; different magnitude; long duration) and sudden fluctuation fault (inconsistent trend; large fluctuation magnitude; short duration), where vehicles No.1, No.4~No.6 have voltage progressive fluctuation fault and No.2, No.3 and No.7 have voltage sudden fluctuation fault. The specific fault information is summarised in Table 2.

The data from vehicles No.1, No.2, No.4, and No.7~No.9 are used as the training dataset for the proposed model. The data from vehicles No.3, No.5 and No.6 are used to analyze the necessity and stability of the proposed method. And No.10 vehicle is used to verify the reliability of the method. In this article, the proposed algorithm is coded with Python 3.7.3, and implemented and tested on a PC (processor 11th Gen Intel (R) Core (TM) i5-11400H @ 2.70 GHz with 8 GB of RAM).

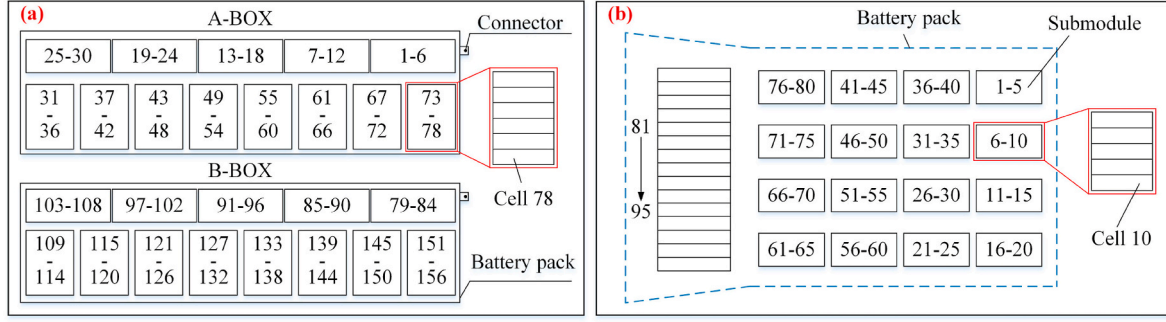


Fig. 2. Battery cell layout diagram. (a) Model 1, (b) model 2.

Table 1

Information of vehicles and batteries.

Vehicle number	Mode	Battery type	Battery cells' number	Number of battery modules	Charge cut-off voltage (V)	Discharge cut-off voltage (V)	Sample frequency (Hz)
Vehicles No.1~No.3, No.9, and No.10	Model 1	NCM	156	26	4.25	3	0.05
Vehicle No.4~No.8	Model 2	NCM	95	17	4.25	3	0.1

3. Diagnosis method of abnormal voltage fluctuation faults

3.1. Extraction of two-dimensional fault characteristics

Construct the retrieved battery voltage data in terms of time and number of individual cells into a voltage matrix U :

$$U = \begin{bmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,N} \\ u_{2,1} & u_{2,2} & \cdots & u_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ u_{T,1} & u_{T,2} & \cdots & u_{T,N} \end{bmatrix} \quad (1)$$

where $u_{t,n}$ denotes the voltage value of the cell n at time t . T denotes the total time. N denotes the total number of cells. $t = 1, 2, \dots, T$. $n = 1, 2, \dots, N$.

3.1.1. Standard deviation

The SD is the most commonly used forms of quantification to reflect the degree of dispersion of a data set, and therefore can be used as a characteristic parameter to characterize voltage fluctuations. Its standard calculation formula is:

$$\sigma = \sqrt{\frac{1}{M} \sum_{m=1}^M (H_m - \mu)^2} \quad (2)$$

where μ and M denote the mean value and the number of elements of the variable H , respectively, and $\mu = \frac{1}{M} \sum_{m=1}^M H_m$.

The SD of all cells under different windows are calculated to obtain the matrix $A_{(L)}$:

$$A_{(L)} = \begin{bmatrix} a_{1,(L)}^1 & a_{2,(L)}^1 & \cdots & a_{N,(L)}^1 \\ a_{1,(L)}^2 & a_{2,(L)}^2 & \cdots & a_{N,(L)}^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,(L)}^C & a_{2,(L)}^C & \cdots & a_{N,(L)}^C \end{bmatrix} \quad (3)$$

where L represents the length of the window. C represents the number of the window, and $C = T-L+1$. $a_{n,(L)}^c$ denotes the SD of the cell n in the c th window with length L . Thus, the SD matrix A at different L can be obtained:

$$A = [A_{(2)}, \dots, A_{(l)}, \dots, A_{(L)}] \quad (1 < l \leq L) \quad (4)$$

To analyze the recognition effect of SD method on two forms of voltage fluctuation faults under different window lengths, it is necessary to calculate the SD matrix B of the fault data in the voltage matrix U under different L . The calculation process is shown in Fig. 4, where i denotes the initial point position of the cell fault and u_i denotes the corresponding voltage value, and j denotes the end point position of the cell fault and u_j is the corresponding voltage value (Note: for TR vehicles, $j + l$ is the initial point position of the cell performance TR). B can be specifically expressed as:

$$B = \begin{bmatrix} B_{(30)}^1 & B_{(31)}^1 & \cdots & B_{(L)}^1 \\ B_{(30)}^2 & B_{(31)}^2 & \cdots & B_{(L)}^2 \\ \vdots & \vdots & \ddots & \vdots \\ B_{(30)}^{c'} & B_{(31)}^{c'} & \cdots & B_{(L)}^{c'} \end{bmatrix} \quad (30 \leq L \leq 300) \quad (5)$$

where $B_{(l)}^{c'} = [b_{1,(l)}^{c'}, \dots, b_{n,(l)}^{c'}, \dots, b_{N,(l)}^{c'}]$, $b_{n,(l)}^{c'} = \sqrt{\frac{1}{l} \sum_{t=i+c'-l}^{t=i+c'-1} (u_{t,n} - \mu_{n,(l)}^{c'})^2}$, $\mu_{n,(l)}^{c'} = \frac{1}{l} \sum_{t=i+c'-l}^{t=i+c'-1} u_{t,n}$. $c' = j - i + l$, $1 \leq c' \leq c$. To minimize noise interference and control calculation costs, $30 \leq L \leq 300$ is taken. Thereby, the recognition effect of abnormal cells under different L can be expressed as:

$$\Delta d = [D_{(30)}, \dots, D_{(l)}, \dots, D_{(L)}] \quad (6)$$

where $D_{(l)}$ denotes the maximum value of the difference between the minimum value of the SD of the faulty cells and the maximum value of the SD of the normal cells in the 1st to c th iterations when the sampling length is l . That is, $D_{(l)} = \max\{d_{(l)}^1, \dots, d_{(l)}^{c'}\}$, $d_{(l)}^{c'} = \min\{b_{fault,(l)}^{c'}\} - \max\{b_{other,(l)}^{c'}\}$, where $b_{fault,(l)}^{c'}$ denotes the SD of the faulty cells in $B_{(l)}^{c'}$, and $b_{other,(l)}^{c'}$ denotes the SD of the other cells in $B_{(l)}^{c'}$ after excluding the faulty cells. When $D_{(l)} > 0$, the SD method can only distinguish abnormal cells from normal cells in the corresponding window l , and the larger $D_{(l)}$ the better the effect.

Based on the above analysis, the variation of Δd with sampling length L for vehicles No.2 and No.4 are calculated as shown in Fig. 5 (a) and (b), respectively, where the initial and end points of the cell fault are $i = j = 310$ for No.2 vehicle and $i = 550$ and $j = 4189$ for No.4 vehicle.

As shown in Fig. 5, the abnormal Cell 131 in No.2 vehicle can be recognized when $L < 185$ and only show a clear differentiation effect

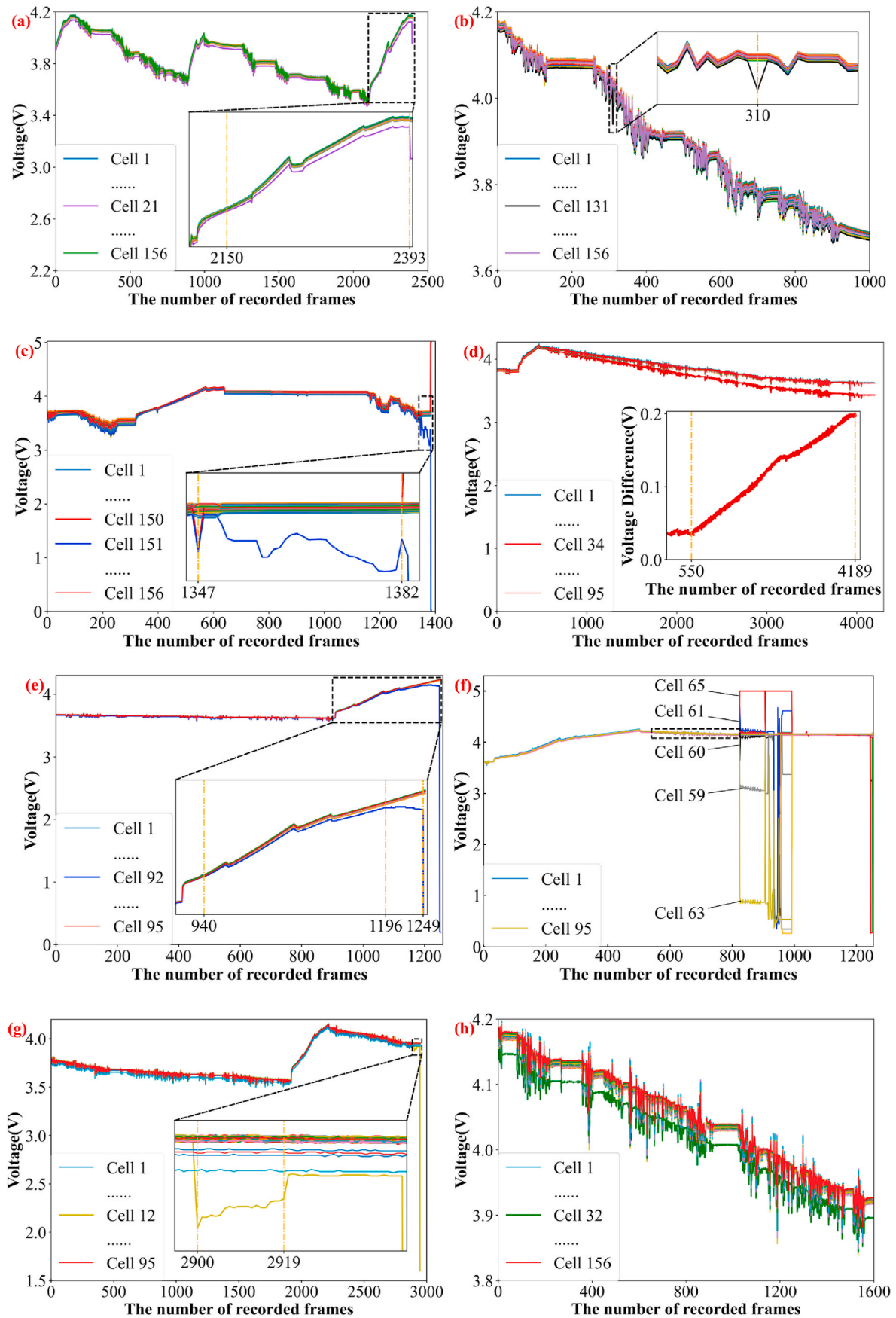


Fig. 3. Cell voltage curve on the day of fault. Vehicles (a) No.1, (b) No.2, (c) No.3, (d) No.4, (e) No.5, (f) No.6, (g) No.7, and (h) No.10.

Table 2
Vehicle fault information.

Vehicle number	Fault categories	Vehicle status	Fault causes
No.1 vehicle	Progressive fault	TR	Short circuit
No.2 vehicle	Sudden fault	Fault alarm	Short circuit
vehicles No.3 and No.7	Sudden fault	TR	Short circuit
No.4 vehicle	Progressive fault	Fault alarm	Poor voltage consistency
vehicle No.5~No.6	Progressive fault	TR	Short circuit
vehicle No.8~No.10	Normal	Normal	/

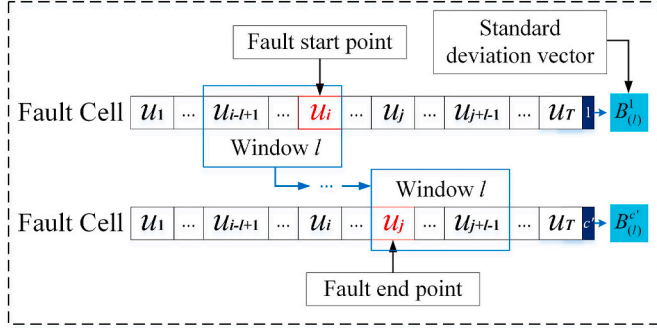


Fig. 4. Schematic diagram of the sliding window of faulty vehicles.

when $L < 125$, and the smaller the window length the better the effect. On the contrary, for No.4 vehicle, the isolation of abnormal cells can only be realized when $L > 154$, and the larger the window length the better the result. Therefore, for vehicles No.2 and No.4 with two different forms of abnormal voltage fluctuation faults, there is no intersection between the sets of time window lengths that effectively identify faulty cells. This indicates that the SD method under a single sliding window of fixed size cannot realize the comprehensive and effective detection of voltage faults.

3.1.2. Correlation

The Pearson correlation coefficient (PCC) is widely used to measure the degree of linear correlation between two variables. In the field of battery fault diagnosis, for series-connected battery packs, all cells have the same current and the voltage fluctuations of each cell are very similar under normal conditions, but when there is an abnormality in the cell, the correlation between the abnormal cells and the normal cells will

become worse, so that the PCC can be used to realize the identification of the abnormal voltage fluctuations. Its standard formula can be expressed as:

$$C_{x,y} = \frac{cov(x,y)}{\sigma_x \cdot \sigma_y} = \frac{\sum_{e=1}^E (x_e - \bar{x})(y_e - \bar{y})}{\sqrt{\sum_{e=1}^E (x_e - \bar{x})^2} \sqrt{\sum_{e=1}^E (y_e - \bar{y})^2}} \quad (7)$$

where $C_{x,y}$ denotes the correlation coefficient of variables x and y . σ_x denotes the SD of variable x . $cov(x,y)$ is the covariance of variables x and y . $\bar{x}$ is the mean of variable x . The correlation coefficient has a very important property and can be expressed by the following equation:

$$\begin{aligned} C_{ax+by} &= \frac{cov(ax+b,y)}{\sigma(ax+b) \cdot \sigma_y} = \frac{\sum_{e=1}^E (ax_e + b - a\bar{x} - b)(y_e - \bar{y})}{\sqrt{\sum_{e=1}^E (ax_e + b - a\bar{x} - b)^2} \sqrt{\sum_{e=1}^E (y_e - \bar{y})^2}} \\ &= \frac{\sum_{e=1}^E (x_e - \bar{x})(y_e - \bar{y})}{\sqrt{\sum_{e=1}^E (x_e - \bar{x})^2} \sqrt{\sum_{e=1}^E (y_e - \bar{y})^2}} = C_{x,y} \end{aligned} \quad (8)$$

where a and b are constants and $a > 0$. Therefore, the correlation of the two variables are independent of their mean and fluctuation magnitude. This mathematical nature of the correlation coefficient dictates that the method is unable to identify voltage progressive fluctuation fault with a consistent trend.

To solve the above problem, this paper proposes an IPCC method, which consists of the following three steps:

- (1) The voltages of all cells are detrended. Specifically, the median value of the voltage matrix U at time t can be expressed as $u_{t(med)}$. The detrended voltage matrix DV can be expressed as:

$$DV = [DV_1, \dots, DV_t, \dots, DV_T]^T \quad (9)$$

where $DV_t = [dv_{t,1}, \dots, dv_{t,n}, \dots, dv_{t,N}]$, and $dv_{t,n} = u_{t,n} - u_{t(med)}$.

- (2) The result of voltage detrending of normal cells is close to 0, thus according to Eq. (10), the noise signal will play a role in determining the size of the correlation coefficient.

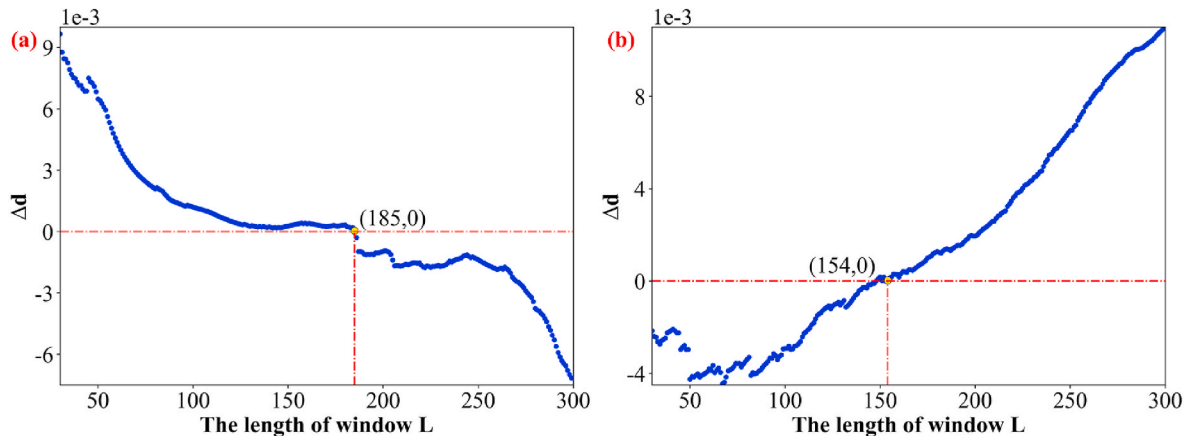


Fig. 5. Identification effect of the faulty cell at different L . (a) Cell 131 of No.2 vehicle, (b) Cell 34 of No.4 vehicle.

$$\begin{aligned}
C_{x+W_1, y+W_2} &= \frac{\text{cov}(x+W_1, y+W_2)}{\sigma(x+W_1) \cdot \sigma(y+W_2)} \\
&= \frac{\sum_{e=1}^E (x_e - \bar{x} + W_{1,e} - \bar{W}_1)(y_e - \bar{y} + W_{2,e} - \bar{W}_2)}{\sqrt{\sum_{e=1}^E (x_e - \bar{x} + W_{1,e} - \bar{W}_1)^2} \sqrt{\sum_{e=1}^E (y_e - \bar{y} + W_{2,e} - \bar{W}_2)^2}} \\
&\approx \frac{\sum_{e=1}^E (W_{1,e} - \bar{W}_1)(W_{2,e} - \bar{W}_2)}{\sqrt{\sum_{e=1}^E (W_{1,e} - \bar{W}_1)^2} \sqrt{\sum_{e=1}^E (W_{2,e} - \bar{W}_2)^2}} = C_{W_1, W_2}
\end{aligned} \quad (10)$$

where W_1 and W_2 are mutually independent noise signals.

To reduce the interference caused by noise signals, discrete square wave periodic signals can be considered to be added as a correction function to the detrended voltage matrix DV . At this time, W_1 and W_2 are the same periodic functions with zero mean and independent of x and y . Thus, the correlation between normal cells after detrending become:

$$C_{x+W_1, y+W_2} \approx \frac{\sum_{e=1}^E (W_{1,e} - \bar{W}_1)(W_{2,e} - \bar{W}_2)}{\sqrt{\sum_{e=1}^E (W_{1,e} - \bar{W}_1)^2} \sqrt{\sum_{e=1}^E (W_{2,e} - \bar{W}_2)^2}} = C_{W_1, W_2} = 1 \quad (11)$$

Therefore, by adding a discrete square wave period signal the correlation between the normal cells can be enhanced. The correction function can be defined as follows:

$$f(t) = \begin{cases} \omega & (0 < t \leq \frac{T_0}{2}) \\ -\omega & (\frac{T_0}{2} < t \leq T_0) \end{cases} \quad (t = 1, 2, \dots, T) \quad (12)$$

where $f(t)$ is a discrete square wave function with amplitude ω and period T_0 , and the magnitude ω depends mainly on the voltage measurement error. $f(t)$ is added to the matrix DV to obtain the new matrix DV' :

$$DV' = \begin{bmatrix} dv_{1,1} + (-1)^\alpha \omega & dv_{1,2} + (-1)^\alpha \omega & \dots & dv_{1,N} + (-1)^\alpha \omega \\ dv_{2,1} + (-1)^\alpha \omega & dv_{2,2} + (-1)^\alpha \omega & \dots & dv_{2,N} + (-1)^\alpha \omega \\ \vdots & \vdots & \ddots & \vdots \\ dv_{T,1} + (-1)^\alpha \omega & dv_{T,2} + (-1)^\alpha \omega & \dots & dv_{T,N} + (-1)^\alpha \omega \end{bmatrix} \quad (13)$$

where $\alpha = 2$ when $0 < t \leq T_0/2$ and $\alpha = 1$ when $T_0/2 < t \leq T_0$.

- (3) It is known that for the early fault of the onboard battery pack, the faulty cells tend to be a minority, and the vast majority of cells in the onboard battery pack are often normal cells, so the median value of each sample can reflect the general trend of all cells, which can be considered as normal cell data. Based on this, the median values of all cells at different times of the matrix DV' are obtained to form the vector $DV'_{med} = [DV'_{1(med)}, \dots, DV'_{t(med)}, \dots, DV'_{T(med)}]^T$, DV'_{med} is chosen as the benchmark for the correlation coefficient calculation, and if the correlation between any cell in matrix DV' and this benchmark is weak, it indicates that the cell has inconsistent voltage fluctuations. Thus, the IPCC for matrix U at L is:

$$R_{(L)} = \begin{bmatrix} r_{1,(L)}^1 & r_{2,(L)}^1 & \dots & r_{N,(L)}^1 \\ r_{1,(L)}^2 & r_{2,(L)}^2 & \dots & r_{N,(L)}^2 \\ \vdots & \vdots & \ddots & \vdots \\ r_{1,(L)}^C & r_{2,(L)}^C & \dots & r_{N,(L)}^C \end{bmatrix} \quad (14)$$

where $r_{n,(L)}^c$ denotes the IPCC between the cell n and the vector DV'_{med} in the c th window of the matrix DV' . Therefore, the IPCC matrix R at different L can be expressed as:

$$R = [R_{(30)}, \dots, R_{(l)}, \dots, R_{(L)}] \quad (30 \leq L \leq 300) \quad (15)$$

Based on the analysis in the previous section, in the same way, the effect function of the IPCC-based method for the identification of anomalous cells under different L can be obtained:

$$\Delta d = [D'_{(30)}, \dots, D'_{(l)}, \dots, D'_{(L)}] \quad (16)$$

where $D'_{(l)}$ denotes the maximum value of the difference between the minimum value of the IPCC of the normal cells and the maximum value of the IPCC of the abnormal cells in the 1st to c th iterations when the sampling length is L . That is, $D'_{(l)} = \max\{d_{(l)}^1, \dots, d_{(l)}^c, \dots, d_{(l)}^C\}$, $d_{(l)}^c = \min\{r_{other,(l)}^c\} - \max\{r_{fault,(l)}^c\}$, where $r_{fault,(l)}^c$ denotes the IPCC of the faulty cells in $R_{(l)}^c$ and $r_{other,(l)}^c$ denotes the IPCC of other cells in $R_{(l)}^c$ except the faulty cells. When $D'_{(l)} > 0$, the IPCC method is able to identify abnormal cells in the corresponding window l , and the larger the $D'_{(l)}$ the better the effect.

Based on the IPCC method, vehicles No.2 and No.4 are taken as examples to calculate their voltage detrending, IPCC and the recognition effect of faulty cells under different L . The results are shown in Fig. 6, where $T_0 = 6$ and $\omega = 0.02$ are taken. From Fig. 6 (a) and (d), it shows that there are noise signals in the detrending results, and if the correlation coefficients of the cells and median vector are calculated directly at this time, results will inevitably be interfered by the noise. However, Fig. 6 (b) and (e) show that the IPCC method can effectively eliminate the interference and thus recognize two forms of voltage faults. From Fig. 6 (c), it can be seen that the IPCC method has higher sensitivity to voltage sudden fault at different L , but still presents the trend that the smaller the window is, the more obvious the effect is. In Fig. 6 (f), for No.4 vehicle, fault isolation can only be realized when $L > 97$, and the larger the window, the better. Therefore, the IPCC method still needs different window lengths to better realize the comprehensive detection of voltage faults.

In summary, both the SD method and the IPCC method can achieve comprehensive and effective detection of different forms of voltage faults under suitable time window lengths, and they have the same adaptive effect on the length of the time window, i.e., a longer time window can better identify voltage progressive fluctuation fault, while the voltage sudden fluctuation fault is more suitable for a shorter time window, and thus we use the SD and IPCC as the two-dimensional fault characteristics characterizing voltage fluctuations, which can achieve a better isolation of faulty cells.

3.2. Assessment method

The extraction of two-dimensional fault features enables quadrant-symmetric isolation of faulty and normal cells. Further, we introduce an improved K-means method to quickly identify the faulty cell set and quantitatively characterize the degree of faults by using the center distance between the clusters of faulty and normal cells as an EC. Before performing K-means clustering we need to complete the following steps (1) and (2):

- (1) Risk assessment of voltage progressive fluctuation fault

The voltage variation rate can more intuitively reflect the state of voltage fluctuation. The voltage variation rate of the cells with progressive fluctuation fault relative to normal cells is expressed as an outlier with a longer duration. The derivative of the cell voltage u can be

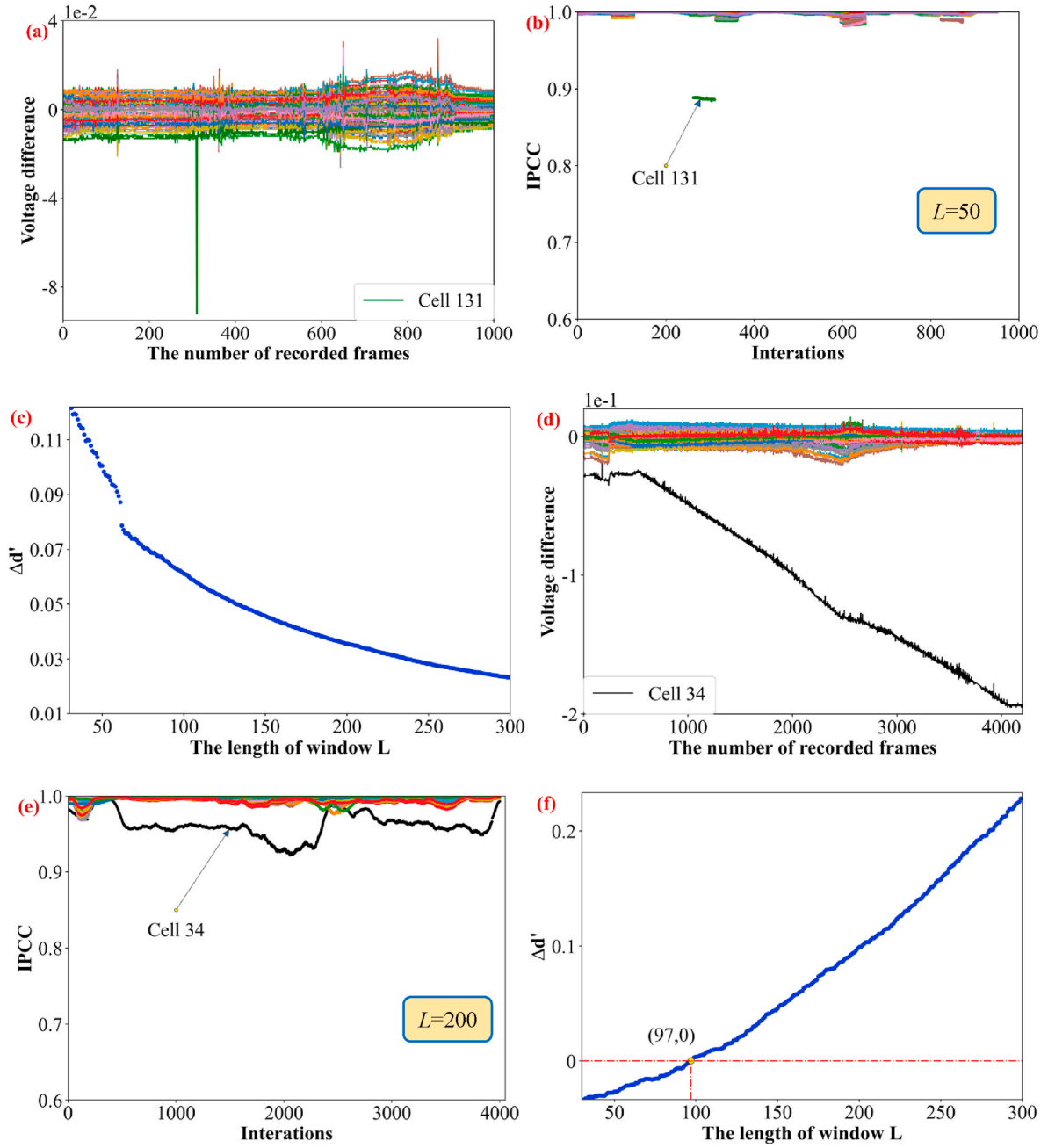


Fig. 6. Cell voltage detrending graph. (a) No.2 vehicle, (d) No.4 vehicle. IPCC. (b) No.2 vehicle at $L = 50$, (e) No.4 vehicle at $L = 200$. Identification of faulty cells at different L . (c) Cell 131 of No.2 vehicle, (f) Cell 34 of No.4 vehicle.

expressed as:

$$\dot{u} = \frac{du}{dt} = \frac{u_{i+1} - u_i}{\Delta t} \quad (17)$$

where u_{i+1} and u_i indicate the cell voltages at the next and current moments, respectively. Δt is a fixed time step interval. Thus, based on the voltage matrix U , the voltage variation rate of all cells in the c th window with length L is:

$$V_{(L)}^c = \begin{bmatrix} v_{1,1}^{c,L} & v_{1,2}^{c,L} & \cdots & v_{1,N}^{c,L} \\ v_{2,1}^{c,L} & v_{2,2}^{c,L} & \cdots & v_{2,N}^{c,L} \\ \vdots & \vdots & v_{g,n}^{c,L} & \vdots \\ v_{G,1}^{c,L} & v_{G,2}^{c,L} & \cdots & v_{G,N}^{c,L} \end{bmatrix} \quad (18)$$

where $v_{g,n}^{c,L}$ indicates the voltage variation rate of the cell n at time g in the c th window of length L , and $v_{g,n}^{c,L} = (u_{g+c,n} - u_{g+c-1,n})/\Delta t$, $G = L-1$.

Since the voltage fluctuations of a faulty cell may be the most violent or the most flat. Therefore, the matrix $V_{(L)}^c$ is normalized and then the absolute value is taken to get the matrix $Z_{(L)}^c$:

$$Z_{(L)}^c = \begin{bmatrix} z_{1,1}^{c,L} & z_{1,2}^{c,L} & \cdots & z_{1,N}^{c,L} \\ z_{2,1}^{c,L} & z_{2,2}^{c,L} & \cdots & z_{2,N}^{c,L} \\ \vdots & \vdots & z_{g,n}^{c,L} & \vdots \\ z_{G,1}^{c,L} & z_{G,2}^{c,L} & \cdots & z_{G,N}^{c,L} \end{bmatrix} \quad (19)$$

where $z_{g,n}^{c,L} = |v_{g,n}^{c,L} - \mu_g^c|/\sigma_g^c$, and μ_g^c and σ_g^c are the mean and SD of the g th

row vector of matrix $V_{(L)}^c$, respectively.

Define the anomaly matrix Q with the same dimension as matrix $Z_{(L)}^c$, and $Q = [Q_1, \dots, Q_g, \dots, Q_G]^T$, $Q_g = [q_{g,1}, \dots, q_{g,n}, \dots, q_{g,N}]$, $q_{g,n}$ is equal to 0 or 1. Locate the position of the maximum value in each row vector in matrix $Z_{(L)}^c$, mark 1 for the same position in Q and 0 for the rest, and then calculate the risk of fault for each cell:

$$P = [p_1, \dots, p_n, \dots, p_N] \quad (20)$$

where p_n represents the fault risk of cell n and $p_n = \frac{1}{G} \sum_{g=1}^{G-G} q_{g,n}$. The cell corresponding to the maximum value in vector P is marked as the cell with the highest fault risk, and this cell is used as the initial clustering center for progressive fault detection at this window.

(2) Risk assessment of voltage sudden fluctuation fault

The voltage variation rate of cells with a sudden fault compared to normal cells takes the form of a severe outlier with a short duration. Therefore, we consider the cell corresponding to the maximum value in matrix $Z_{(L)}^c$ in Eq. (19) as the object with the highest risk of fault in this window, and use this cell as the initial center of clustering for sudden fluctuation fault detection.

(3) Improved K-means method

To eliminate the quantitative differences between the two-dimensional fault features, we normalize the SD and IPCC of each calculation and then cluster the results. The major procedure of conventional K-means clustering algorithm is shown in Table 1 of the Appendix.

The initialization centroid of K-means is random, and different initialization centroid has a large impact on the result of the algorithm

and the computational cost. Therefore, to optimize the K-means algorithm, we follow steps (1) and (2) to find the initial clustering centers of the K-means method when we perform the identification of voltage progressive fault and sudden fault, respectively. When there is a fault in the battery pack, all cells will be classified into two categories, so K is equal to 2 by default in this paper, and whether there are faulty cells in the classification result we judge by the Euclidean distance between two clustering centers:

$$\eta = \text{euc}(\varphi_1, \varphi_2) \quad (21)$$

where φ_1 and φ_2 represent the two cluster centers, and η is EC. If $\eta > \eta_\delta$, it indicates that there is an abnormal voltage fluctuation fault in the battery pack, where the cluster with the smaller sample will be identified as the set of all faulty cells, thus the number and location of faulty cells are obtained. On the contrary, if $\eta < \eta_\delta$, it is considered normal for all cells. η_δ is a threshold of the EC.

In summary, the flowchart of the proposed fault diagnosis strategy in this paper is shown in Fig. 7.

4. Selection of key parameters

The diagnostic results need to be determined by the window length L and the EC threshold η_δ , so this section focuses on the selection of reasonable parameters L and η_δ . The selection process is shown in Fig. 8 and consists of three steps.

The first step is a pre-selection layer for the window length L . Accident vehicle data with abnormal voltage fluctuation faults are selected. Based on the SD method, the recognition effect function Δd for the faulty cells in vehicle F_v can be calculated according to Eqs. (1)–(6) at different L , and then the set $L_{SD}^{F_v}$ of valid windows can be expressed as:

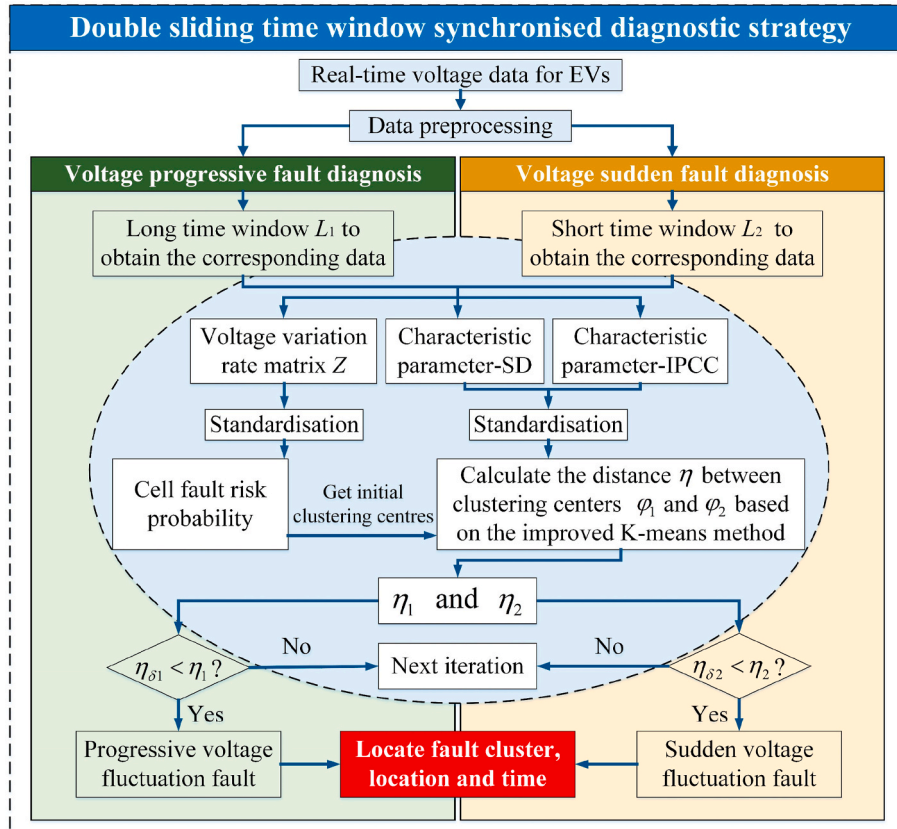


Fig. 7. Flowchart of fault detection scheme.

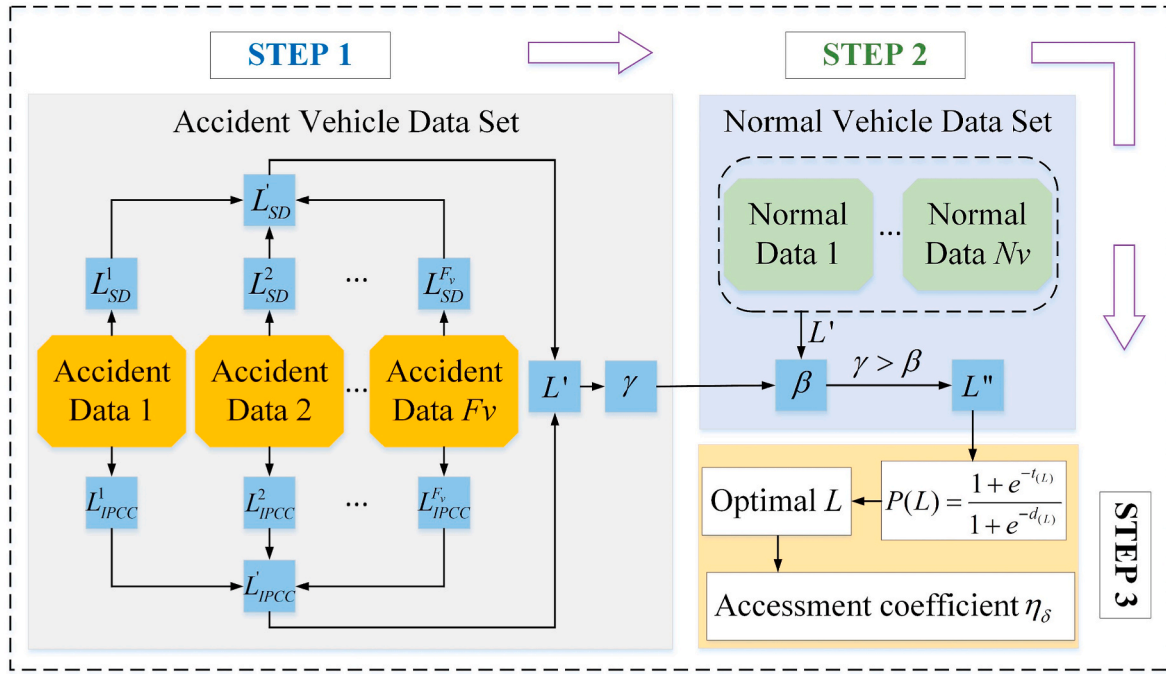


Fig. 8. Flowchart for the selection of key parameters.

$$L_{SD}^{Fv} = \begin{cases} \{L\}, & \text{if } D_{(L)} > 0 \\ \emptyset, & \text{if } D_{(L)} < 0 \end{cases} \quad (22)$$

Similarly, further combining Eqs. (7)–(16), the set L_{IPCC}^{Fv} of windows that can effectively identify faulty cells can be obtained based on the IPCC method.

Therefore, the common set of valid window lengths extracted from all faulty vehicles:

$$L' = L_{SD}^{Fv} \cap L_{IPCC}^{Fv} = L_{SD}^1 \cap L_{SD}^2 \cap \dots \cap L_{SD}^{Fv} \cap L_{IPCC}^1 \cap L_{IPCC}^2 \cap \dots \cap L_{IPCC}^{Fv} \quad (23)$$

Further, combining Eq. (21), the evaluation coefficients (ECs) matrix for the faulty data segment of vehicle F_v at different L is calculated as follows:

$$\eta^{Fv} = \begin{bmatrix} \eta_{(30)}^{1,Fv} & \eta_{(31)}^{1,Fv} & \dots & \eta_{(L)}^{1,Fv} \\ \eta_{(30)}^{2,Fv} & \eta_{(31)}^{2,Fv} & \dots & \eta_{(L)}^{2,Fv} \\ \vdots & \vdots & \ddots & \vdots \\ \eta_{(30)}^{c,Fv} & \eta_{(31)}^{c,Fv} & \dots & \eta_{(L)}^{c,Fv} \end{bmatrix} \quad (L \in L') \quad (24)$$

where $\eta_{(l)}^{c,Fv}$ denotes the EC of the fault data of vehicle F_v in the c th window of length L . Thus, the minimum η value of different L for vehicle F_v can be expressed as: $\gamma^{Fv} = [\gamma_{(30)}^{Fv}, \dots, \gamma_{(l)}^{Fv}, \dots, \gamma_{(L)}^{Fv}]$, where $\gamma_{(l)}^{Fv} = \min\{\eta_{(l)}^{1,Fv}, \dots, \eta_{(l)}^{c,Fv}, \dots, \eta_{(l)}^{Fv}\}$. It is worth noting that, in the calculation of $\eta_{(l)}^{c,Fv}$, both $d_{(l)}^{c,Fv}$ and $d_{(l)}^{Fv}$ in the fault cells identification effect function of the two-dimensional fault characteristics should be greater than 0 to ensure that the η value at this iteration has a fault isolation effect, otherwise $\eta_{(l)}^{c,Fv}$ is rejected. Thus, for different faulty vehicles, the minimum η value under different L can be expressed as:

$$\gamma = [\gamma_{(30)}, \dots, \gamma_{(l)}, \dots, \gamma_{(L)}] \quad (L \in L') \quad (25)$$

where $\gamma_{(l)} = \min\{\gamma_{(l)}^1, \gamma_{(l)}^2, \dots, \gamma_{(l)}^{Fv}\}$.

The second step is the second level of filtering for the window L . Normal vehicle data and normal data for faulty vehicles before voltage

fluctuation faults are obtained. Based on the voltage matrix U of normal vehicles N_v , the ECs for different L are calculated:

$$\eta^{Nv} = \begin{bmatrix} \eta_{(30)}^{1,Nv} & \eta_{(31)}^{1,Nv} & \dots & \eta_{(L)}^{1,Nv} \\ \eta_{(30)}^{2,Nv} & \eta_{(31)}^{2,Nv} & \dots & \eta_{(L)}^{2,Nv} \\ \vdots & \vdots & \ddots & \vdots \\ \eta_{(30)}^{c,Nv} & \eta_{(31)}^{c,Nv} & \dots & \eta_{(L)}^{c,Nv} \end{bmatrix} \quad (L \in L') \quad (26)$$

where $\eta_{(l)}^{c,Nv}$ denotes the EC of vehicle N_v in the c th window with length L . For vehicle N_v , the maximum η value for different L can be expressed as: $\beta^{Nv} = [\beta_{(30)}^{Nv}, \dots, \beta_{(l)}^{Nv}, \dots, \beta_{(L)}^{Nv}]$, where $\beta_{(l)}^{Nv} = \max\{\eta_{(l)}^{1,Nv}, \dots, \eta_{(l)}^{c,Nv}, \dots, \eta_{(l)}^{Nv}\}$. Thereby, for different normal vehicles, the maximum η value for different L can be expressed as:

$$\beta = [\beta_{(30)}, \dots, \beta_{(l)}, \dots, \beta_{(L)}] \quad (L \in L') \quad (27)$$

where $\beta_{(l)} = \max\{\beta_{(l)}^1, \beta_{(l)}^2, \dots, \beta_{(l)}^{Nv}\}$. In order to avoid misdiagnosis of normal cells, the ECs of the faulty cell have the largest value when normal data and faulty data take the same window L . L' that meets the condition can be obtained by

$$L'' = \begin{cases} \{L\}, & \text{if } \gamma_{(L)} > \beta_{(L)} \quad \& \quad L \in L' \\ \emptyset, & \text{if } \gamma_{(L)} \leq \beta_{(L)} \quad \& \quad L \in L' \end{cases} \quad (28)$$

The third step is the selection layer for the optimal window length and the EC threshold. The results of the selection of key parameters should be able to distinguish faulty cells from normal ones, while the computational cost should be as low as possible. In this paper, the comprehensive performance assessment function $P(L)$ of the model is defined as follows according to the Sigmoid function:

$$P(L) = \frac{1 + e^{-t(L)}}{1 + e^{-d(L)}} \quad (L \in L'') \quad (29)$$

where $t(L)$ indicates the calculation time of a single sliding window of length L and $d(L)$ indicates the difference between $\gamma_{(L)}$ and $\beta_{(L)}$, i.e. $d(L) = \gamma_{(L)} - \beta_{(L)}$. When $d(L)$ is larger and $t(L)$ is smaller, $P(L)$ is larger

and the comprehensive performance of the model is better. When $P(L)$ takes the maximum value, the corresponding time window length L is optimal, and the EC threshold η_δ should satisfy $\beta_{(L)} < \eta_\delta < \gamma_{(L)}$.

Based on the three steps above, voltage data from faulty vehicles No.1 and No.4 and normal vehicles No.8 and No.9 are used to determine the window length and the EC threshold for detecting voltage progressive fluctuation fault. Where the initial and end points of the fault for No.1 vehicle are $i = 2150$ and $j = 2393$, respectively. The effective window length is determined to be in the range $154 < L \leq 300$ according to Eq. (23). Fig. 9 (a) illustrates the variation of the distance between the two clustering centers for two faulty vehicles under different L , where the black dots are the values of the minimum EC γ for faulty vehicles. Fig. 9 (b) illustrates the variation of the distance between the two clustering centers with different L calculated from the normal data of the

two normal vehicles and the four faulty vehicles, and the black dots are the values of the maximum EC β taken for the normal vehicles. Fig. 9 (c) illustrates the results calculated according to Eq. (29), which shows that $P(L)$ has a maximum value of 1.12 when $L = 239$, corresponding to a minimum EC of 5.7 for faulty vehicles and a maximum EC of 3.54 for all normal vehicles. Therefore, the window length is taken to be $L = 239$ and the EC threshold η_δ to be 5 for the detection of voltage progressive fluctuation fault.

Similarly, the faulty vehicles No.2 and No.7 and the normal data of the faulty vehicles are used to determine the optimal window length and EC threshold for the sudden voltage fluctuation fault. Similarly, the effective window lengths are determined to be in the range $30 \leq L < 185$ according to Eq. (23). The results are shown in Fig. 9 (d), (e), and (f). The maximum EC β calculated from the normal data in Fig. 9 (e) is divided

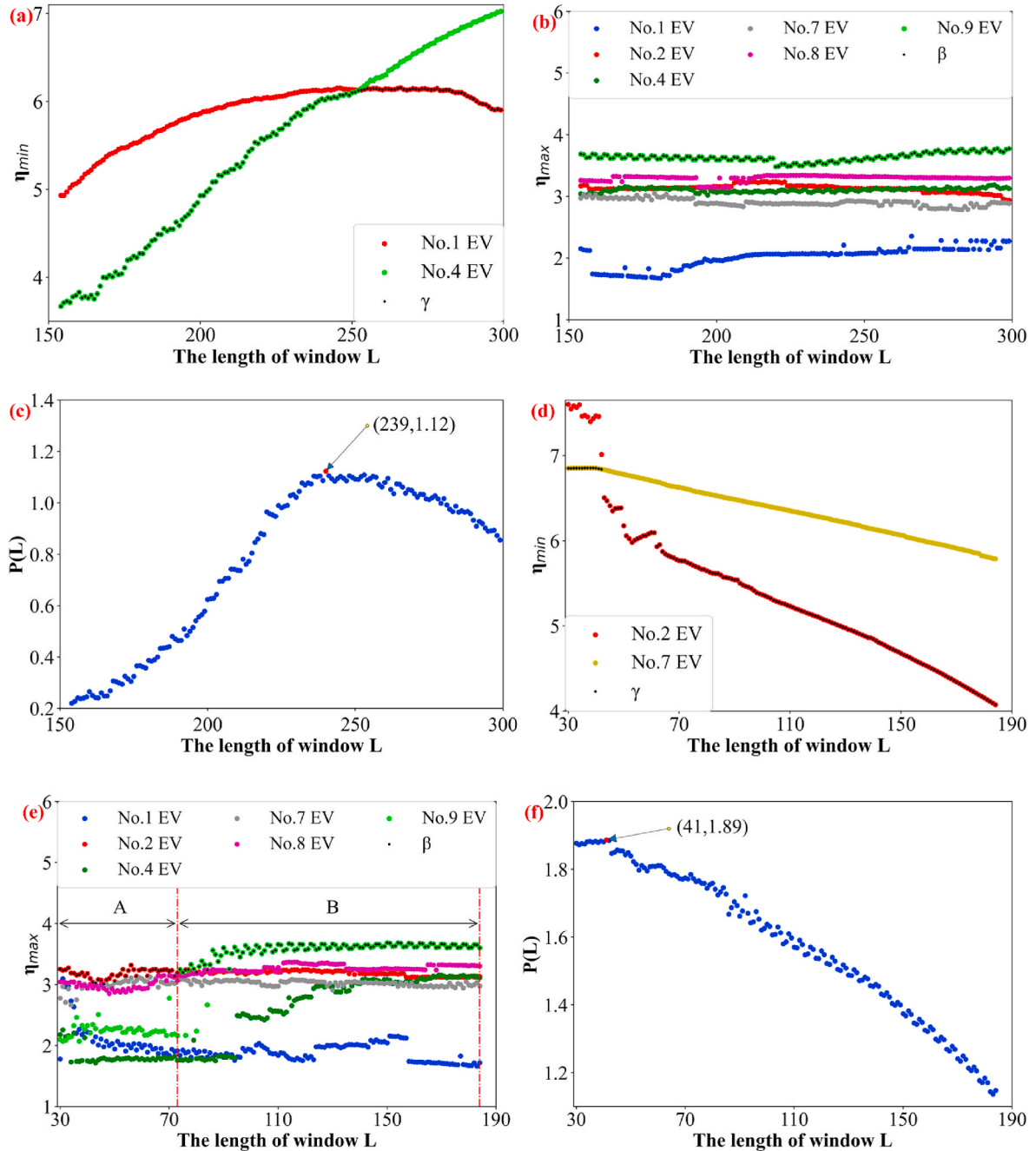


Fig. 9. ECs. (a) Vehicles No.1 and No.4, (d) vehicles No.2 and No.7, (b) normal data for six EVs when $154 < L \leq 300$ and (e) $30 \leq L < 185$. Relationship between $P(L)$ and L . (c) Progressive fault, (f) sudden fault.

into two stages (stage A and B) from No.2 and No.9 vehicles, respectively. Fig. 9 (f) shows that $P(L)$ has a maximum value of 1.89 when $L = 41$, corresponding to a minimum EC of 6.85 for faulty vehicles and a maximum EC of 3.10 for normal vehicles. Therefore, the sliding window length L is taken as 41 and the EC threshold η_δ as 6 when detecting sudden voltage fluctuation fault.

5. Validation of the method

5.1. Necessity analysis

In this section, we perform a verification of the necessity of the dual sliding time window synchronization implementation strategy based on the voltage data of the No.3 vehicle with sudden fluctuation fault and the No.5 vehicle with progressive fluctuation fault.

Based on the analysis in Section 4, the voltage data of the two vehicles are calculated according to the algorithm flow proposed in this paper, and the results are shown in Fig. 10. For No.3 vehicle, Fig. 10 (a) and (b) show the ECs at $L = 41$ and $L = 239$, respectively. Where Fig. 10 (a) shows that the ECs detect a fault from the 1307th iteration. Combined with Fig. 3 (c), it can be seen that the window of this calculation just contains the first frame data of the sudden fault (frame 1347) of this vehicle, and thus the method has a very high sensitivity to the sudden voltage fluctuation fault at $L = 41$. However, when $L = 239$, the ECs should detect the fault for the first time at the 1109th (1347-239+1=1109) iteration, but shows in Fig. 10 (b), there is a serious fault detection delay (17 iterations). Therefore, a short time window can be more effective in improving the sensitivity of sudden fluctuation fault detection.

For No.5 vehicle, Fig. 3 (e) shows that the voltage data from approximately the 940th frame to the 1196th frame of this vehicle is the early stage of the progressive fault, and the fault characteristic is not

obvious, and the maximum voltage difference only increases from 13mV to 33mV. From the 1196th frame to 1249th frame, the abnormal fluctuation of the faulty cell starts to intensify significantly, and the maximum voltage difference increases to 105mV. Fig. 10 (c) shows that the fault is first detected at the 1168th iteration when $L = 41$, and the latest data in this window is the 1208th (1168+41-1=1208) frame. Therefore, the ECs corresponding to this window length are not sensitive enough to the initial progressive fault, and the fault identification can only be realized when the fault characteristics are obvious enough. In Fig. 10 (d), when $L = 239$, the fault can be detected in the 818th iteration, and the latest data in this window is the 1056th (818+239-1=1056) frame, and the battery is in the earliest stage of the fault. Therefore, a long time window can more effectively improve the sensitivity of progressive fluctuation fault detection. Therefore, in order to improve the sensitivity of the voltage fault identification, it is necessary to implement the dual sliding window strategy for two different forms of voltage faults.

In addition to the above example, based on the analysis in Section 3.1, we know that when $L = 41$, the short window length is unable to capture the characteristics of a voltage progressive fluctuation fault with a long duration and a low degree of fault, and thus the ECs of the training sample No.4 vehicle do not exceed the threshold until the fault alarm. In addition, when $L = 239$, the excessively long window length is highly influenced by the historical data, and the features of brief and weak sudden faults are easily masked, and thus the ECs of the training sample No.2 vehicle are always below the threshold. Therefore, in order to achieve the comprehensiveness and effectiveness of fault detection, it is equally necessary to adopt the dual sliding time window synchronization implementation strategy for different fault characteristics. At the same time, a reasonable value for the window length L should be considered for both the accuracy of diagnosis result and the timeliness of fault detection.

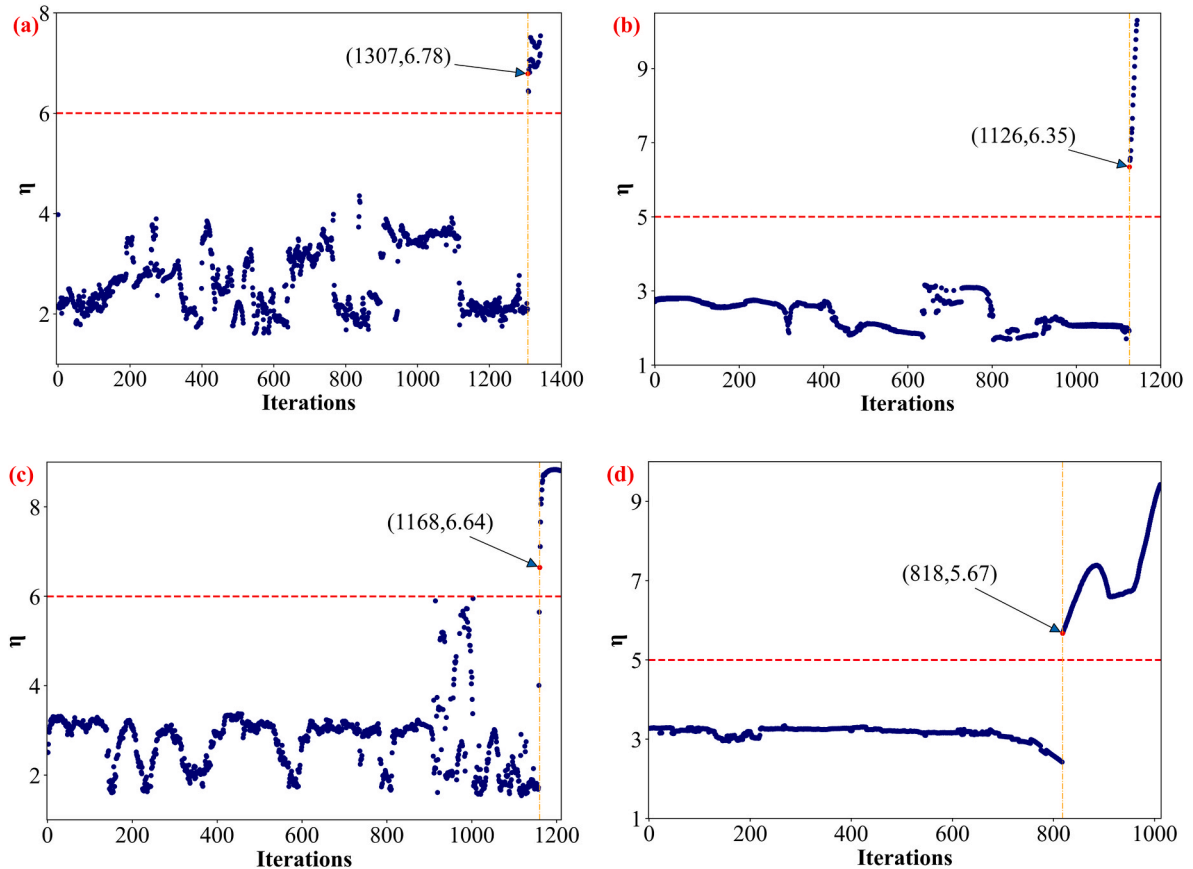


Fig. 10. No.3 vehicle. (a) ECs of $L = 41$, (b) ECs of $L = 239$. No.5 vehicle. (c) ECs of $L = 41$, (d) ECs of $L = 239$.

5.2. Reliability analysis

In order to validate the reliability of the method proposed in this paper, the voltage data of normal No.10 vehicle is selected for analysis in this section.

The ECs are calculated for No.10 vehicle with $L = 41$ and $L = 239$, respectively, and the results are shown in Fig. 11. As can be seen from the figure when $L = 41$ and $L = 239$, the ECs never exceed the threshold. Therefore, for normal vehicles, the fault diagnosis strategy in this paper does not produce false alarms for normal cells and is not affected by cell voltage inconsistencies of the type shown in Fig. 3 (h), so the diagnosis results have a high reliability.

5.3. Stability analysis

The retrieved voltage data of the accident vehicle No.6 is used to analyze the stability of the method in this paper. The vehicle suffered from abnormal voltage fluctuations in Cell 6, Cell 17, Cell 29, Cell 41, Cell 53, Cell 65, Cell 76, Cell 86 and Cell 91 prior to TR, with Cell 65 being the most severe. For the voltage data in the black dashed box in Fig. 3 (f) no any abnormality can be seen by the naked eye alone, but it shows the form of the voltage fluctuations as shown in Fig. 12 (a), it shows that there are significant low voltage fluctuations in Cell 65. A separate comparison of the voltage variation curves for Cell 65 and the normal cells in Fig. 12 (b), and it shows that there are alternating maximum and minimum values for the faulty Cell 65, and the other eight cells show the same abnormal form as Cell 65, only the fluctuation amplitude is smaller. Fig. 12 (c) and (d) show the ECs of No.6 vehicle before TR at $L = 41$ and $L = 239$, respectively. Due to the large voltage fluctuation and long duration of the faulty cells, the ECs of both window lengths can realize the fault identification. But, the short window length can detect faults up to 90 iterations earlier than the long one, so the implementation of the dual sliding window lengths can mutually compensate for the delay in fault identification and thus achieve the earliest possible fault diagnosis. Fig. 12 (e) indicates the fault risk probability of each cell in the 500th iteration at $L = 239$, where Cell 65 has a maximum value of 0.276 and hence will be considered as the initial center of this clustering. The two-dimensional fault characteristics of each cell in this iteration after normalisation are shown in Fig. 12 (f). The two cluster centers are φ_1 and φ_2 , respectively, and the distance between them is 5.21 greater than the threshold of 5, where φ_1 has a smaller number of clusters and is judged to be a faulty cell set. The faulty cells can then be obtained as Cell 6, Cell 17, Cell 29, Cell 41, Cell 53, Cell 65, Cell 76, Cell 86, and Cell 91. Thus, the method can accurately identify multi-cell faults with stable fault diagnosis results.

5.4. Comparative analysis of different methods

Based on the voltage data of faulty vehicles No.2, No.4, and No.6, the method proposed in this paper is analyzed in comparison with the Shannon entropy method [23] and the correlation coefficient method [26]. A confusion matrix is introduced to quantitatively assess the accuracy of the different methods:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (30)$$

where ACC is the accuracy rate. TP represents the number of correctly identified faulty cells and FP represents the number of falsely identified faulty cells. TN represents the number of correctly identified normal cells and FN represents the number of falsely identified normal cells.

The charging and discharging cut-off voltages and the maximum voltage difference before the fault of the three vehicles do not exceed the threshold, and the conventional threshold-based method cannot detect the voltage fault within the safe range. Fig. 13 (a), (d), and (g) shows the ECs of the proposed method. The number of iterations is 960, 3962, and 784, respectively, and the window L is selected as 41, 239, and 41. The fault start time is frame 310, frame 550, and frame 545, respectively, and the end time is frame 310, frame 4189, and frame 823. Combined with the vehicle voltage data, it can be obtained that the number of ECs exceeding the threshold for No.2, No.4, and No.6 vehicles should be 41 ($310-310+41=41$), 3640 ($4189-550+1=3640$), and 279 ($823-545+1=279$), respectively. The actual number of faults detected for each vehicle in the figure is 41, 3544, and 270, respectively, and therefore the number of FP that did not correctly recognize the faulty samples is 0, 186, and 9, respectively. Other than that, the rest of the iterations accurately differentiate between the normal and faulty cells. Thus, the accuracy of the method in this paper is obtained as 100%, 97.6%, and 98.9%, respectively.

The diagnostic results based on Shannon entropy are shown in Fig. 13 (b), (e), and (h). The window length L is set to 100 and the threshold is ± 4 . The number of iterations is 901, 4042, and 725, respectively. Similarly, we can obtain the number of falsely identified faulty samples FP is 95, 3385, and 185, respectively. From the figure, the number of TP is 5, 255, and 94, respectively. The number of FN is 11, 318, and 107. Thus, we can get the ACC of No.2, No.4, and No.6 vehicles, which are 88.2%, 8.4%, and 59.7%. Through the validation of real-world vehicle data, we found that the false positive rate of the method based on Shannon entropy is higher, which is mainly due to the fact that the method is based on intervals to calculate the probability, and the intervals change with the voltage data resulting in a more randomized method, which leads to a higher number of false alarms and lower sensitivity to faults.

The diagnostic results based on the correlation coefficient are shown

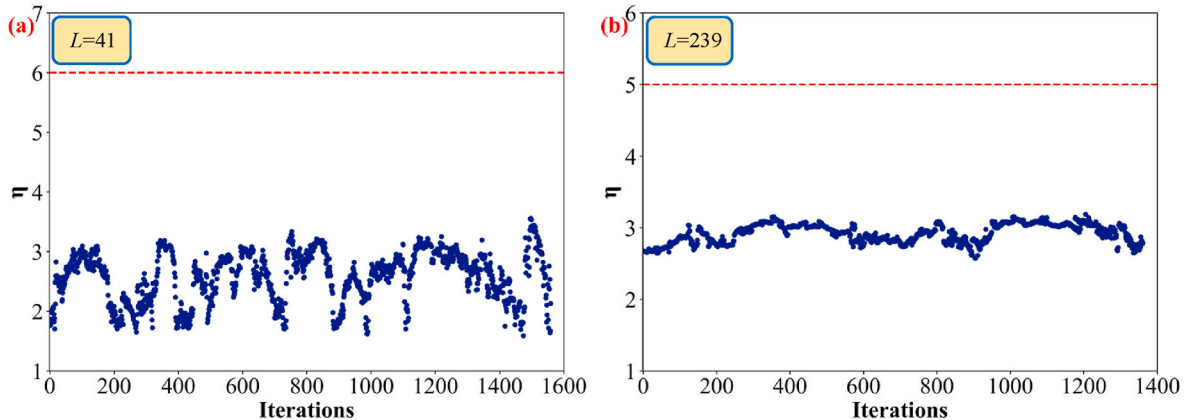


Fig. 11. ECs of No.10 vehicle. (a) $L = 41$, (b) $L = 239$.

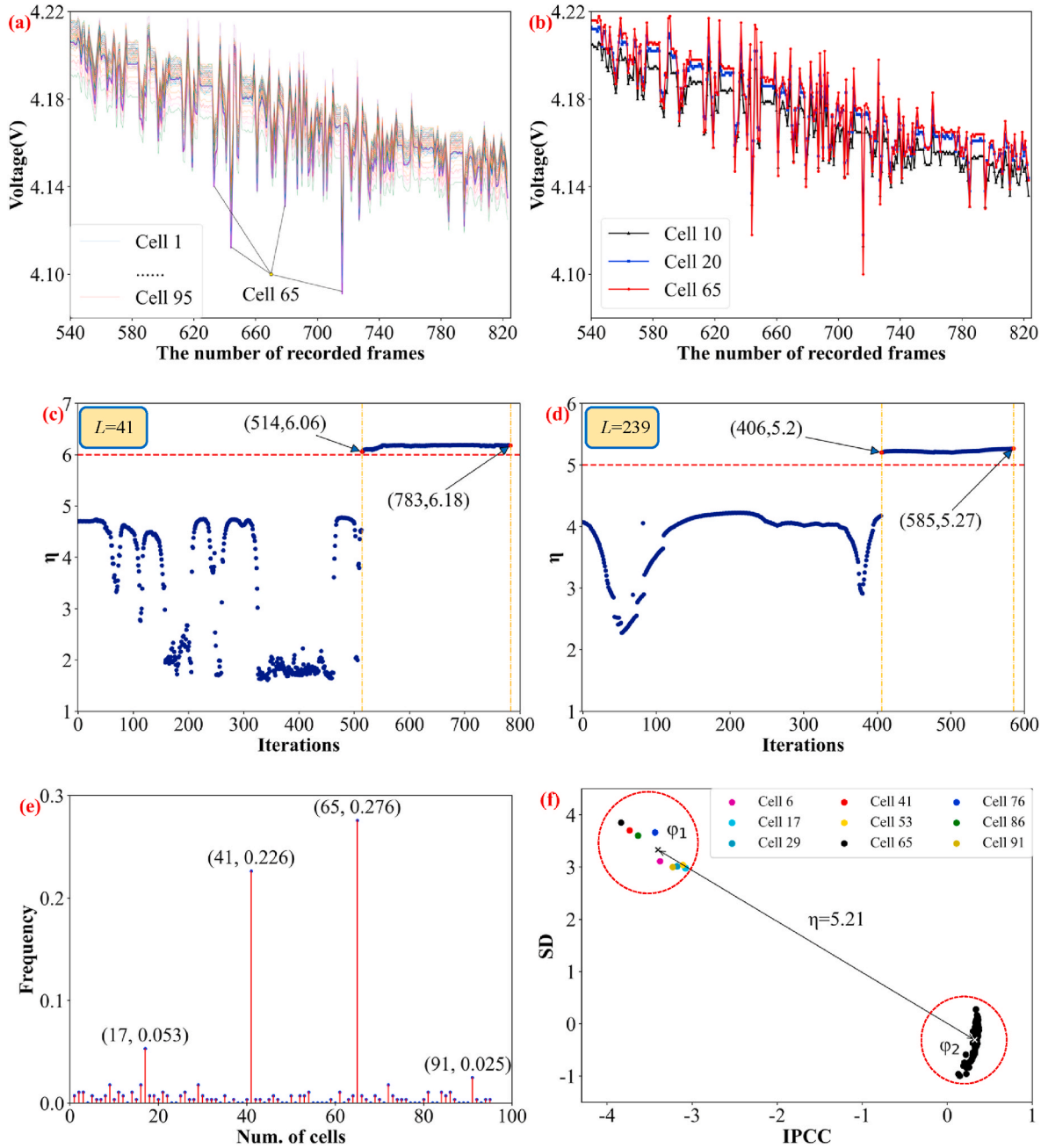


Fig. 12. (a) Voltage of all cells 1h before TR, (b) Cell 65 voltage curve comparison, (c) ECs of $L = 41$, (d) ECs of $L = 239$, (e) probability of abnormal risk for each cell and (f) the IPCC and SD at 500th iteration when $L = 239$.

in Fig. 13 (c), (f), and (i). The window length is chosen to be 30 samples and the threshold to be 0.9. Thus the number of iterations is 971, 4171, and 793, respectively. The FP, TP and FN for No.2 vehicle is 0, 30, and 10, respectively, resulting in an ACC of 98.9% for No.2 vehicle. However, Vehicles No.4 and No.6 should have detected persistent faults after the 521th and 516th iterations, respectively, but Fig. 13 (f) and (i) show that both vehicles have a TP of 0 and a TN of 491 and 493, respectively. Thus, ACCs of No.4 and No.6 vehicles are 11.8% and 62.2%. The reason for the low accuracy of the correlation coefficient method is mainly due to the mathematical properties of the method which leads to its insensitivity to voltage progressive fluctuation fault, and the reason for the misdiagnosis may be caused by the sampling length and accuracy.

In a word, the proposed method has better ACC than the Shannon entropy and the PCC method in Table 3. Although it takes longer time to calculate, its calculation time of a single window is still far less than the

sampling period of real-world vehicle data. In Fig. 12 (f), when a single-step calculation with $L = 239$ is performed, based on the improved K-means method in this paper, the initial clustering center can be obtained in advance as Cell 65, and thus the centroid can be found only costs 0.021s with 2-step calculations, while the traditional method needs 0.049s with 6-step calculations. Therefore, the proposed method greatly reduces the computational cost. Meanwhile, during the simultaneous implementation of double sliding windows, the average time of single calculation based on the method of this paper is only 0.034s, which ensures the comprehensiveness and accuracy of fault detection without sacrificing the computational efficiency, and has the conditions for on-line application.

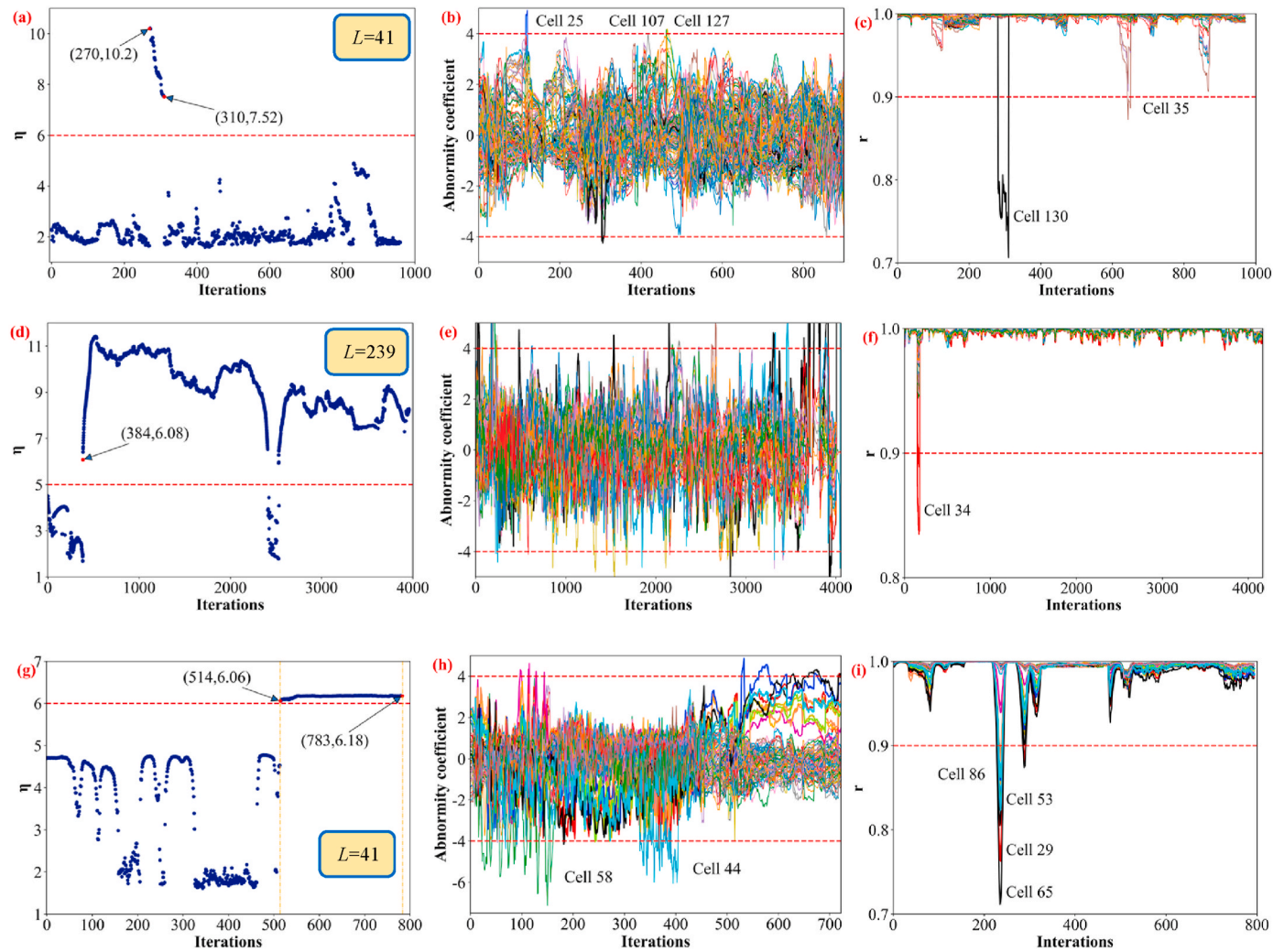


Fig. 13. ECs of the proposed method of (a) No.2, (d) No.4, and (g) No.6 vehicles. The Shannon entropy of (b) No.2, (e) No.4, and (h) No.6 vehicles. The improved PCC of (a) No.2, (d) No.4, and (g) No.6 vehicles.

Table 3
Comparison of detection results for different methods.

Method	ACC			Average calculation time(s) of single window		
	No.2 vehicle	No.4 vehicle	No.6 vehicle	No.2 vehicle	No.4 vehicle	No.6 vehicle
Traditional K-means method	100%	97.6%	98.9%	0.064	0.082	0.054
The proposed method				0.036	0.038	0.029
Modified Shannon entropy	88.2%	8.4%	59.7%	0.028	0.021	0.017
Improved PCC	98.9%	11.8%	62.2%	0.023	0.019	0.014

6. Conclusions

In this paper, an efficient and accurate online diagnosis and prediction method for voltage inconsistency fluctuation faults is proposed. Firstly, the SD and the IPCC are extracted as the two-dimensional fault features characterizing the voltage fluctuation, in which the IPCC method makes up for the defect of the traditional PCC method that is unable to detect the progressive voltage fluctuation fault. Then, based

on the analysis of real-world vehicle data, we found that the two-dimensional fault features have the same window length adaptation effect on voltage fluctuation, i.e., a longer window can better recognize the voltage progressive fluctuation fault, while the voltage sudden fluctuation fault is more suitable for a shorter window. Further, we utilize the K-means method for anomalous cells identification and fault degree assessment. In order to optimize the K-means method, the probability of abnormal fluctuation of each cell in the window is evaluated by the characteristic parameter of voltage variation rate, so as to determine the initial centroid of this clustering, which can effectively reduce the number of clustering iterations and improve the computational efficiency. In order to realize the online application of the method and the comprehensive and effective detection of different forms of voltage fluctuation faults, this paper proposes for the first time the simultaneous implementation strategy of the double sliding time window, and optimal window lengths and evaluation coefficient thresholds are found based on the data-driven method. Finally, the necessity, reliability and stability of this paper's method are verified, and the results indicate the proposed method could identify TR hazards ahead of time. Compared with the improved Shannon entropy method and the improved correlation coefficient method, the fault diagnosis results of this paper's method are more reliable and stable, and the average accuracy rate is more than 98%.

Finally, in addition to the study in this paper, in later work we will further analyze the properties exhibited by this paper's method when

both progressive and sudden faults occur at the same time, and find a suitable method to differentiate between normal cells, progressive faults cells and sudden faults cells, for example, by setting K equal to 3 in the K-means algorithm.

CRedit authorship contribution statement

Qiquan Liu: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Jian Ma:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Xuan Zhao:** Conceptualization, Funding acquisition, Resources, Supervision, Editing. **Kai Zhang:** Data curation, Software, Writing – original draft. **Dean Meng:** Software, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix

Table 1

Steps of the common K-means algorithm

Number	Specific operation
Step 1	Assuming the data are $\{H_1, H_2, \dots, H_m, \dots, H_M\}$, $H_m \in R^n$.
Step 2	The cluster centers $\{\varphi_1, \varphi_2, \dots, \varphi_h\}$ are randomly selected, where h is a positive integer.
Step 3	The Euclidean distance between the sample H_m and each cluster center φ is calculated. The sample H_m is classified to the cluster center with the minimum distance principle: $c^{(m)} := \arg\min_h \ H_m - \varphi_h\ ^2$.
Step 4	For category h , its cluster center will be recalculated: $\varphi_h := \frac{\sum_{m=1}^M \{c^{(m)} = h\} H_m}{\sum_{m=1}^M \{c^{(m)} = h\}}$.
Step 5	The process from Steps 2 to 4 is executed cyclically until the cluster center does not change.

References

- [1] Wang CY, Zhang G, Ge S, Xu T, Ji Y, Yang XG, et al. Lithium-ion battery structure that self-heats at low temperatures. *Nature* 2016;529:515–8. <https://doi.org/10.1038/nature16502>.
- [2] Hannan MA, Lipu MSH, Hussain A, Mohamed A. A review of lithium-ion battery state of charge estimation and management system in electric vehicle applications: challenges and recommendations. *Renew Sustain Energy Rev* 2017;78:834–54. <https://doi.org/10.1016/j.rser.2017.05.001>.
- [3] Wu C, Zhu C, Ge Y, Zhao Y. A review on fault mechanism and diagnosis approach for Li-ion batteries. *J Nanomater* 2015;2015. <https://doi.org/10.1155/2015/631263>.
- [4] Rahimi-Eichi H, Ojha U, Baronti F, Chow MY. Battery management system: an overview of its application in the smart grid and electric vehicles. *IEEE Ind Electron Mag* 2013;7:4–16. <https://doi.org/10.1109/MIE.2013.2250351>.
- [5] Li F, Min Y, Zhang Y. A novel method for lithium-ion battery fault diagnosis of electric vehicle based on real-time voltage. *Wireless Commun Mobile Comput* 2022;2022. <https://doi.org/10.1155/2022/7277446>.
- [6] Li Z, Yang Y, Li L, Wang D. A weighted Pearson correlation coefficient based multi-fault comprehensive diagnosis for battery circuits. *J Energy Storage* 2023;60: 106584. <https://doi.org/10.1016/j.est.2022.106584>.
- [7] Wang G, Zhao J, Yang J, Jiao J, Xie J, Feng F. Multivariate statistical analysis based cross voltage correlation method for internal short-circuit and sensor faults diagnosis of lithium-ion battery system. *J Energy Storage* 2023;62:106978. <https://doi.org/10.1016/j.est.2023.106978>.
- [8] Zhang C, Zhao S, Yang Z, He Y. A multi-fault diagnosis method for lithium-ion battery pack using curvilinear Manhattan distance evaluation and voltage difference analysis. *J Energy Storage* 2023;67:107575. <https://doi.org/10.1016/j.est.2023.107575>.
- [9] Cai L, Wang H, Dong Z, He Z, Gao M, Song Y. A multi-fault diagnostic method based on category-reinforced domain adaptation network for series-connected battery packs. *J Energy Storage* 2023;60:106690. <https://doi.org/10.1016/j.est.2023.106690>.
- [10] Lee SH, Kim J, Kim BH, Yoon S, Cho KY. Delamination-free multifunctional separator for long-term stability of lithium-ion batteries. *Small* 2019;15. <https://doi.org/10.1002/smll.201804980>.
- [11] Luo W, Zhou L, Fu K, Yang Z, Wan J, Manno M, et al. A thermally conductive separator for stable Li metal anodes. *Nano Lett* 2015;15:6149–54. <https://doi.org/10.1021/acs.nanolett.5b02432>.
- [12] Schmitt J, Raatz A. Failure mode based design and optimization of the electrode packaging process for large scale battery cells. *Adv Mater Res* 2014;907:309–19. <https://dx.doi.org/10.4028/www.scientific.net/AMR.907.309>.
- [13] Li D, Zhang Z, Liu P, Wang Z, Zhang L. Battery Fault diagnosis for electric vehicles based on voltage abnormality by combining the long short-term memory neural network and the equivalent circuit model. *IEEE Trans Power Electron* 2021;36: 1303–15. <https://doi.org/10.1109/TPEL.2020.3008194>.
- [14] Feng X, Pan Y, He X, Wang L, Ouyang M. Detecting the internal short circuit in large-format lithium-ion battery using model-based fault-diagnosis algorithm. *J Energy Storage* 2018;18:26–39. <https://doi.org/10.1016/j.est.2018.04.020>.
- [15] Chen Z, Xiong R, Tian J, Shang X, Lu J. Model-based fault diagnosis approach on external short circuit of lithium-ion battery used in electric vehicles. *Appl Energy* 2016;184:365–74. <https://doi.org/10.1016/j.apenergy.2016.10.026>.
- [16] Ma M, Wang Y, Duan Q, Wu T, Sun J, Wang Q. Fault detection of the connection of lithium-ion power batteries in series for electric vehicles based on statistical analysis. *Energy* 2018;164:745–56. <https://doi.org/10.1016/j.energy.2018.09.047>.
- [17] Zhao Y, Liu P, Wang Z, Zhang L, Hong J. Fault and defect diagnosis of battery for electric vehicles based on big data analysis methods. *Appl Energy* 2017;207: 354–62. <https://doi.org/10.1016/j.apenergy.2017.05.139>.
- [18] Sun Z, Han Y, Wang Z, Chen Y, Liu P, Qin Z, et al. Detection of voltage fault in the battery system of electric vehicles using statistical analysis. *Appl Energy* 2022;307: 118172. <https://doi.org/10.1016/j.apenergy.2021.118172>.
- [19] Li D, Zhang Z, Liu P, Wang Z. DBSCAN-based thermal runaway diagnosis. 2019.
- [20] Chen Z, Xu K, Wei J, Dong G. Voltage fault detection for lithium-ion battery pack using local outlier factor. *Meas J Int Meas Confed* 2019;146:544–56. <https://doi.org/10.1016/j.measurement.2019.06.052>.
- [21] Yao L, Fang Z, Xiao Y, Hou J, Fu Z. An intelligent fault diagnosis method for lithium battery systems based on grid search support vector machine. *Energy* 2021; 214:118866. <https://doi.org/10.1016/j.energy.2020.118866>.
- [22] Sun J, Qiu Y, Shang Y, Lu G. A multi-fault advanced diagnosis method based on sparse data observers for lithium-ion batteries. *J Energy Storage* 2022;50:104694. <https://doi.org/10.1016/j.est.2022.104694>.
- [23] Wang Z, Hong J, Liu P, Zhang L. Voltage fault diagnosis and prognosis of battery systems based on entropy and Z-score for electric vehicles. *Appl Energy* 2017;196: 289–302. <https://doi.org/10.1016/j.apenergy.2016.12.143>.
- [24] Shang Y, Lu G, Kang Y, Zhou Z, Duan B, Zhang C. A multi-fault diagnosis method based on modified Sample Entropy for lithium-ion battery strings. *J Power Sources* 2020;446:227275. <https://doi.org/10.1016/j.jpowsour.2019.227275>.

- [25] Hong J, Wang Z, Chen W, Wang L. Multi-fault synergistic diagnosis of battery systems based on the modified multi-scale entropy. *Int J Energy Res* 2019;43: 8350–69. <https://doi.org/10.1002/er.4831>.
- [26] Xia B, Shang Y, Nguyen T, Mi C. A correlation based fault detection method for short circuits in battery packs. *J Power Sources* 2017;337:1–10. <https://doi.org/10.1016/j.jpowsour.2016.11.007>.
- [27] Kang Y, Duan B, Zhou Z, Shang Y, Zhang C. A multi-fault diagnostic method based on an interleaved voltage measurement topology for series connected battery packs. *J Power Sources* 2019;417:132–44. <https://doi.org/10.1016/j.jpowsour.2019.01.058>.
- [28] Kang Y, Duan B, Zhou Z, Shang Y, Zhang C. Online multi-fault detection and diagnosis for battery packs in electric vehicles. *Appl Energy* 2020;259:114170. <https://doi.org/10.1016/j.apenergy.2019.114170>.